AI-Enabled Operational & Logistics Transformation at NAC Kazatomprom JSC

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Kazatomprom should run three AI pilots, with people keeping control of safety and finance

Pilot SCADA telemetry and AI acid dosing at Inkai, adapt AI freight tracking on the TITR, and pilot one robotic packing arm at Ulba, while emergency shutdowns, groundwater protection and financial checks stay human-controlled [1], [7].

1

Inkai: 90-day SCADA pilot

Proposed target: 5%–6% less acid per kg U

2

TITR: AI freight tracking

Proposed target: 10 days shorter transit

3

Ulba: drum-packing pilot

Proposed target: 30% fewer exposure hours

Acid supply, corridor delays and locked-up inventory limit Kazatomprom more than demand does

- Sulfuric acid shortage led to lower 2025 production expectations [8]
- About 48% of Western shipments moved to the TITR (Assignment 1 evidence); adds transfers, Caspian weather and customs [9]
- KZT 423,314M inventory at end-2023, KZT 328,015M of it uranium products [1]
- Result: the binding limits are chemistry, transit time and cash, not sales

Sources: [1], [8], [9]

The state owner wants domestic investment, and the Subsoil Code limits how fast AI can change operations

Stakeholder	What they expect	Effect on my design
Samruk-Kazyna (75%)	Dividends, capital kept in Kazakhstan, regional jobs	Reject foreign M&A; retrain staff
Public investors (25%)	Returns, clear reporting (London, Astana)	Named reviewer for all financial figures
Ministry of Energy	Subsoil Code groundwater protection	Shutdowns and escalations stay human-only

Sources: [1], [7]

Protect low-cost ISR extraction and add software, not foreign reactor assets

- Kazatomprom: 100% in-situ recovery; acid solution through sandy ground, so no ore is moved [6]
- Cameco: underground hard-rock mining (McArthur River, Cigar Lake) and a 49% Westinghouse stake [3]
- Different mining method means only process software and logistics practices can transfer
- Foreign reactor M&A conflicts with the 75% state owner’s domestic focus [1]

Sources: [1], [3], [6]

Reject two Cameco ideas, adopt or pilot two, and adapt one

Cameco idea	Verdict	Why
49% Westinghouse-style M&A [3]	Reject	Conflicts with state capital goals [1]
Underground mining automation [3]	Reject	ISR has no hard-rock mines [6]
Automated process control, telemetry	Adopt / Pilot	Fits ISR; targets acid use [6]
Automated drum packing	Pilot	One arm at Ulba
Shipment tracking after Inkai delay [3]	Adapt	Redesign for multi-border TITR [9]

Sources: [3], [6], [9]

AI screens and drafts; named people decide on safety, law and money

Tier	Wellfield	Ulba	TITR	Finance
Human only	Shutdowns, groundwater	Safety, licence	Contract talks	Sign statements
Human + AI	Reserve modelling	Pack scheduling	Inventory swaps	Scenarios
AI + human review	Dosing alerts; Lead Engineer signs above ±5%	Fault alerts; supervisor	ETA variance >24 h; Logistics Officer	Ratios; Senior Analyst
Automated	Telemetry, 20,000+ wellheads	Drum filling	Status data	Data matching
Fallback	>3% drift: 6-h manual rounds	Manual tele-manipulator	Manual customs filing	Manual check

Sources: [7], [10] (proposed design)

Continuous telemetry would replace a 48–72 hour manual lag (design assumption)

	Current state (assumed)	Proposed 90-day pilot, one Inkai block
Sampling	Manual rounds, 20,000+ wellheads	Continuous SCADA downhole telemetry
Delay	48–72 h lab results	Near real time
Dosing	Late valve changes	AI drafts; Lead Engineer approves shifts above ±5%
Escalation	—	>12 bar alerts Environmental Safety Director
Fallback	—	Drift >3%: manual 6-hour rounds
Target	—	5%–6% less acid per kg U (proposed)

Sources: [6]; parameters proposed in Assignments 1 and 2

Utilities buy reliability, so tracking and buffers matter as much as price

- Cameco 2023: only two-thirds of an Inkai shipment arrived in Q4; the rest came later at a Canadian port. Managed with inventory buffers, long-term contracts and loans [3]
- Kazatomprom has an ANU Energy inventory loan arrangement [1]
- Proposed TITR tracking: AI predicts ETA; Logistics Officer checks variance >24 h; port closure >72 h goes to Commercial Director
- Manual customs filing if the data feed fails
- Proposed target: 10 days shorter European transit [9]

Sources: [1], [3], [9]

Staff need judgement, adaptability and the skill to check AI

Capability	Example at Kazatomprom	Who is trained
Human	Ethical judgement on delays and groundwater risk	Wellfield engineers
Meta	Adapting when a route closes	Logistics staff
Digital	Checking AI ETAs, dosing alerts and ratios	Engineers, junior analysts

WEF 2025 [4] and OECD 2024 [5] discuss shifting skill demand; they are general evidence, not Kazatomprom data.

Sources: [4], [5]

Every AI action has a named approver, a limit and a manual fallback

Risk	Control	Owner
Privacy, security	Data stays in company systems; role-based approval	IT Security Lead
Route bias	Check recommendations against alternate corridors	Logistics Officer
Telemetry reliability	Drift >3% → manual 6-hour hydrometer rounds	Lead Engineer
Accountability	Action log: who approved what	Operations Manager
Safety	>12 bar → Environmental Safety Director halts block	Env. Safety Director
Customs data loss	Manual customs filing	Logistics Officer

Sources: [7] (proposed controls)

Profit is strong, but only KZT 211,912M is cash against KZT 423,314M in inventory

Audited 2023 (KZT million)	Value	Audited 2023	Value
Revenue	1,434,635	Current liabilities	430,812
Net profit	580,335	Current ratio	3.00
Net profit margin	40.45%	Inventory (uranium products)	423,314 (328,015)
Current assets	1,290,336	Cash and equivalents	211,912
Operating cash flow	432,225	Cash ratio	0.49

Sources: [1], [2]

Only the 2023 figures are verified; pilot KPIs are proposed targets

Item	Status	Basis
2023 financials and ratios	Verified	Audited statements [1], [2]
Ownership; Cameco facts	Verified	[1], [3]
48–72 h lag; 20,000+ wellheads; ±5%; 12 bar; 3%; 6 h	Design assumption	Assignments 1 and 2
5%–6% less acid; −10 days; −30% exposure	Pilot target	Benchmarked on [6], [3]
Pilot costs and KZT savings	Not estimated	No verified data

Sources: [1], [2], [3], [6]

A 12-month plan with owners, gates and fallbacks

Action	Owner	KPI (baseline → target)	Dependency	Cost basis	Risk + control	Gate
M0–3: SCADA telemetry + ±5% dosing gate, one Inkai block	Inkai Ops Manager; Lead Engineer	Lag 48–72 h → near real time; acid → 5%–6% lower	Sensors, IT	Pilot sensors; quote needed	Drift; manual rounds	M3: scale or stop
M4–8: TITR tracking; Ulba robot pilot	Logistics Officer; Plant Safety Mgr	Transit → −10 days; exposure → −30%	Data access; safety approval	Licence, robot; quotes needed	API outage, robot fault; manual fallbacks	M8: extend or stop
M9–12: scale to more blocks; train staff; quarterly ratio checks	CTO; HR; Senior Analyst	Blocks 1 → agreed; zero unreconciled ratios	Pilot results	Training time	Over-trust in AI; review gate	M12: board decision

Sources: Proposed plan; targets are pilot targets

The model fits together: the plan runs only inside governance and cash limits



- §4 sets who approves; §6 sets the limits that §5 actions must stay within
- §7 shows cash is tight (0.49), so §8 gates each step on pilot results
- Any failed gate shrinks scope; it never removes human review

I corrected an unverified 4.02 current ratio to 3.00 and declared AI use

Check	Early draft	Audited result
Current assets	KZT 1,250.0bn (unverified)	KZT 1,290,336M
Current liabilities	KZT 310.5bn (unverified)	KZT 430,812M
Current ratio	4.02	3.00
Cash ratio	not calculated	0.49

Level 3 AI use: ChatGPT (brainstorming), Gemini (cross-checking, mapping), Claude (website and drafting of report and deck). I checked all figures against [1], [2] and made the verdicts myself.

Sources: [1]

If a shock hits, scope shrinks but human review stays

Shock	Still holds	Changes	Abandoned
(a) Sensor drift or failure	Human shutdown rule; 12 bar escalation	Block returns to 6-h manual rounds	Automated dosing on that block until trusted
(b) TITR corridor or customs shock	24 h variance check; 72 h escalation	Use inventory buffers [1]; manual customs	10-day target for that period
(c) Engineer shortage	Human-only tier; ±5% gate	One block only; training earlier	Scale-up in Months 9–12

Sources: [1], [7]

References

[1] NAC Kazatomprom JSC, Integrated Annual Report 2023, Astana, 2024.

[2] NAC Kazatomprom JSC, “Financial results of NAC Kazatomprom JSC for 2023,” Astana, 2024.

[3] Cameco Corporation, 2023 Annual Report (Form 40-F / Ex. 99.1), Saskatoon, 2024.

[4] World Economic Forum, The Future of Jobs Report 2025, Geneva, 2025.

[5] A. Green, “AI and the changing demand for skills,” OECD AI Papers, no. 14, Apr. 2024.

[6] IAEA, Manual of Acid In Situ Leach Uranium Mining Technology, IAEA-TECDOC-1239, Vienna, 2001.

[7] Republic of Kazakhstan, Code on Subsoil and Subsoil Use (No. 125-VI), Dec. 27, 2017.

[8] World Nuclear News, “Kazatomprom lowers 2025 uranium production expectations,” Aug. 2024.

[9] The Astana Times, “Kazakhstan Increases Uranium Shipments via TITR,” May 2024.

[10] G. Yeshenkulova et al., Business Administration, Almaty: ZIALY BASPASY, 2026; OpenStax, Introduction to Business, 2nd ed., 2026.